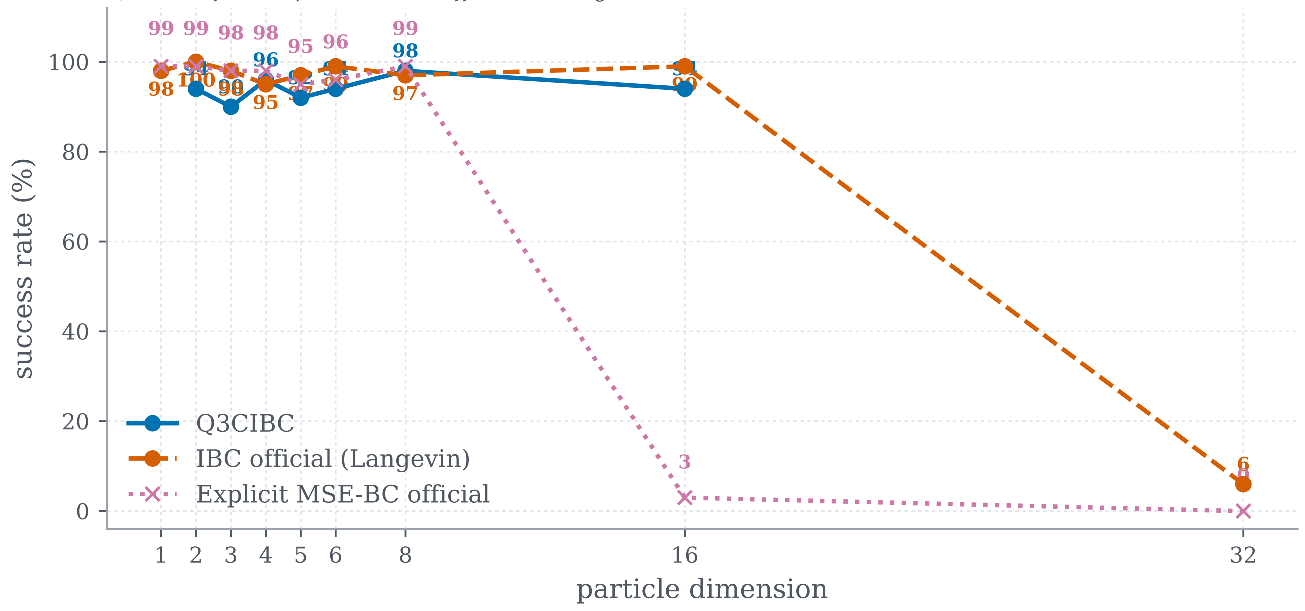
Quality–latency trade-offs across environments

Final comparison results; connected lines denote variants of the same method (door excluded).

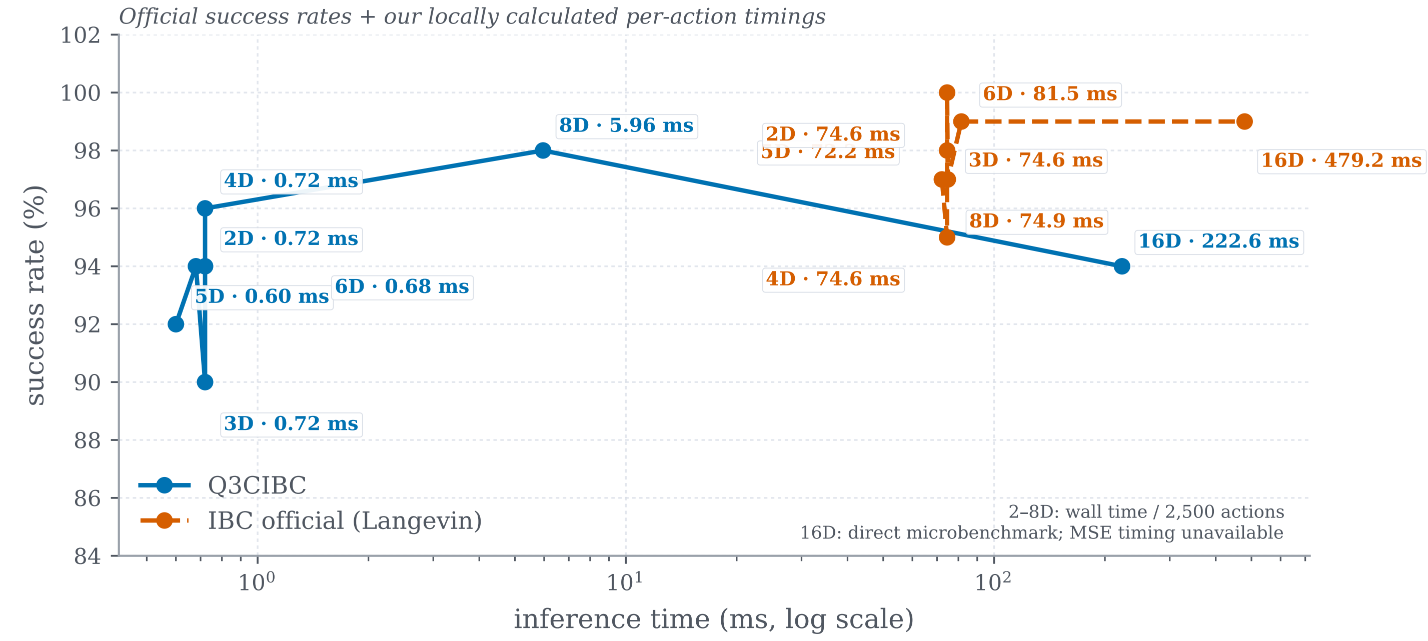
(a) Particle: success vs. dimension

Q3C best final experiments vs official IBC Figure 6 curves



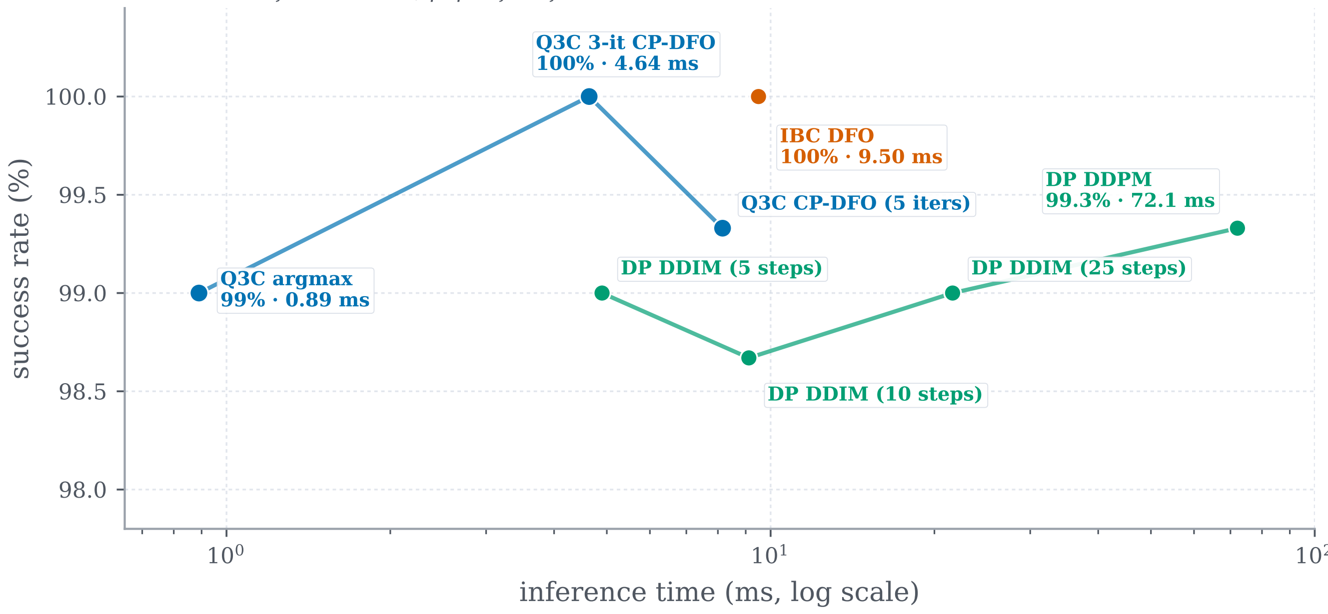
(b) Particle: success vs. inference time

Official success rates + our locally calculated per-action timings



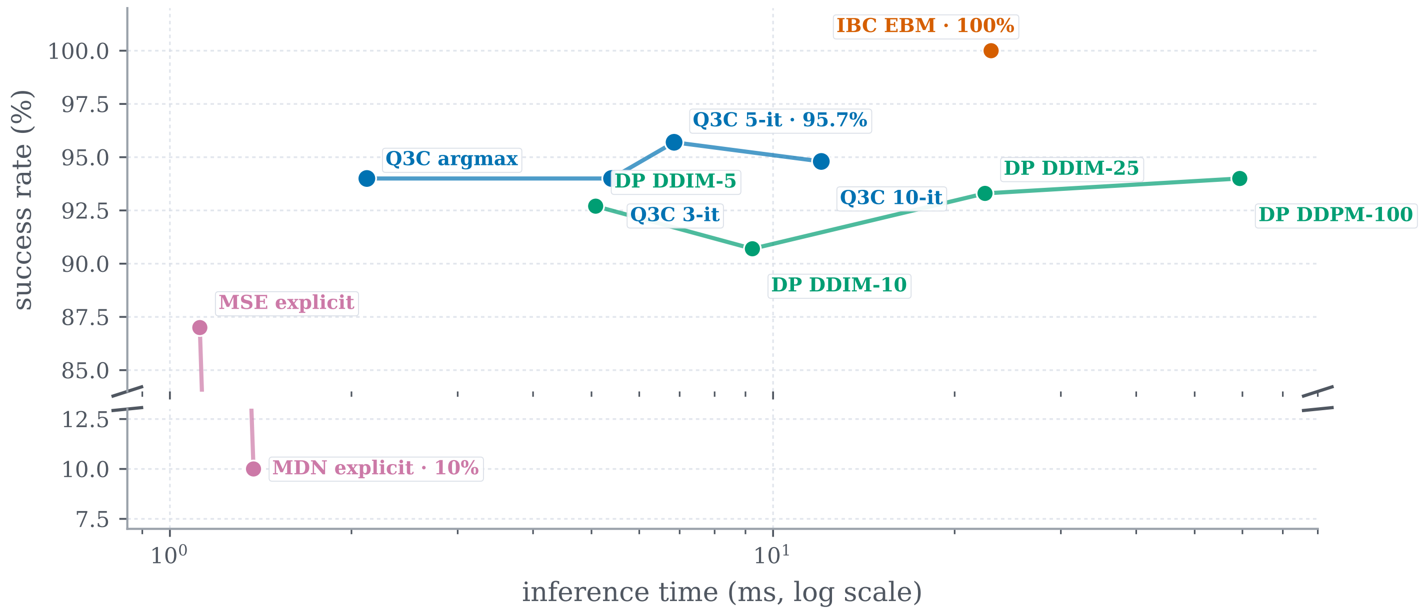
(c) Pushing (states)

Success vs. inference time; paper-faithful architecture



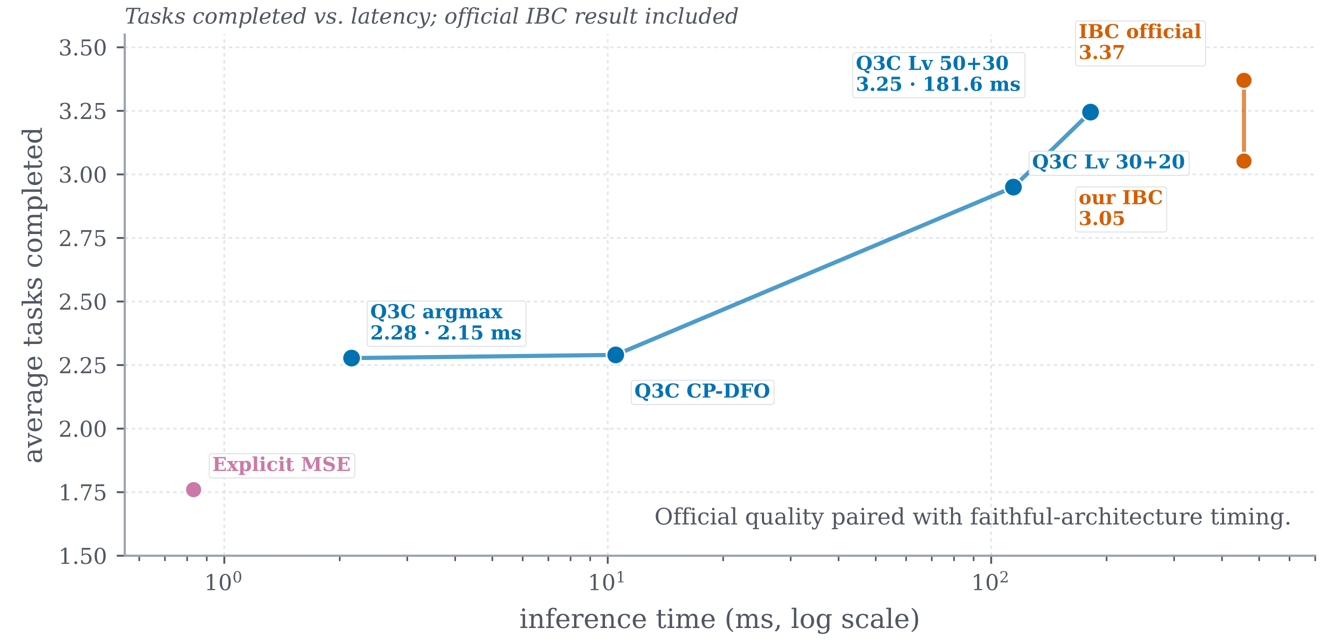
(d) Pushing (pixels)

Broken success axis resolves the 87–100% cluster



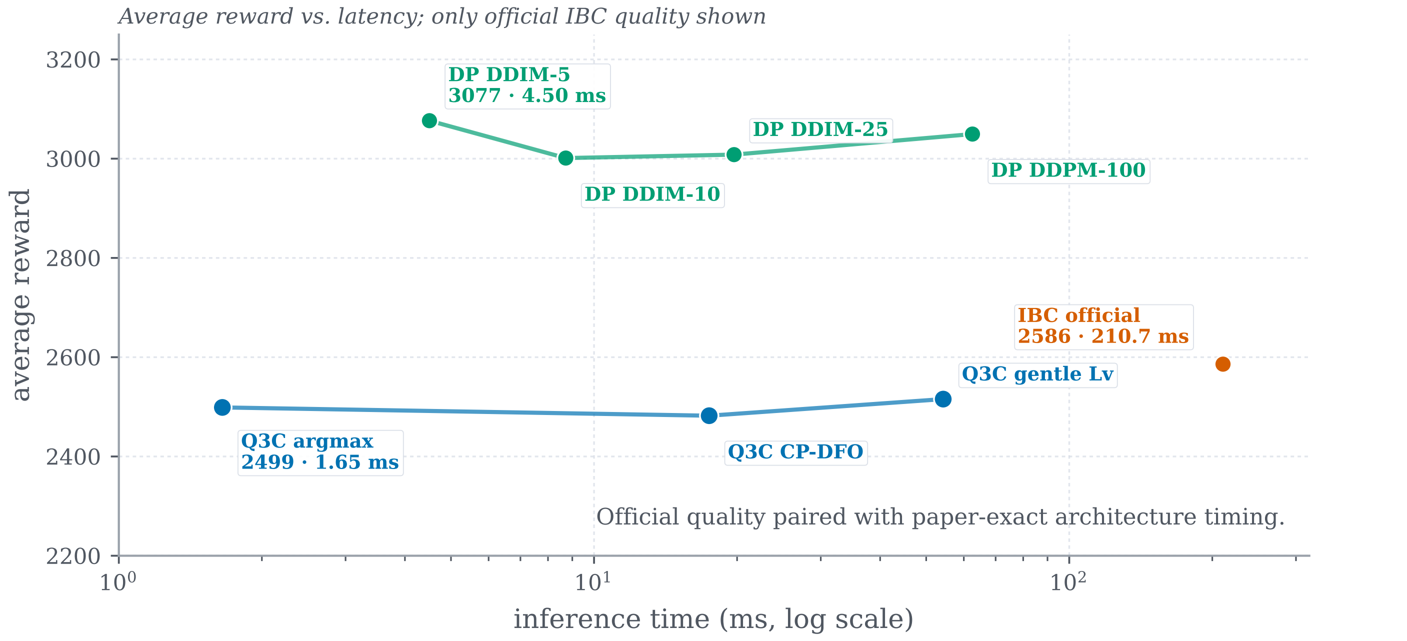
(e) D4RL Kitchen

Tasks completed vs. latency; official IBC result included



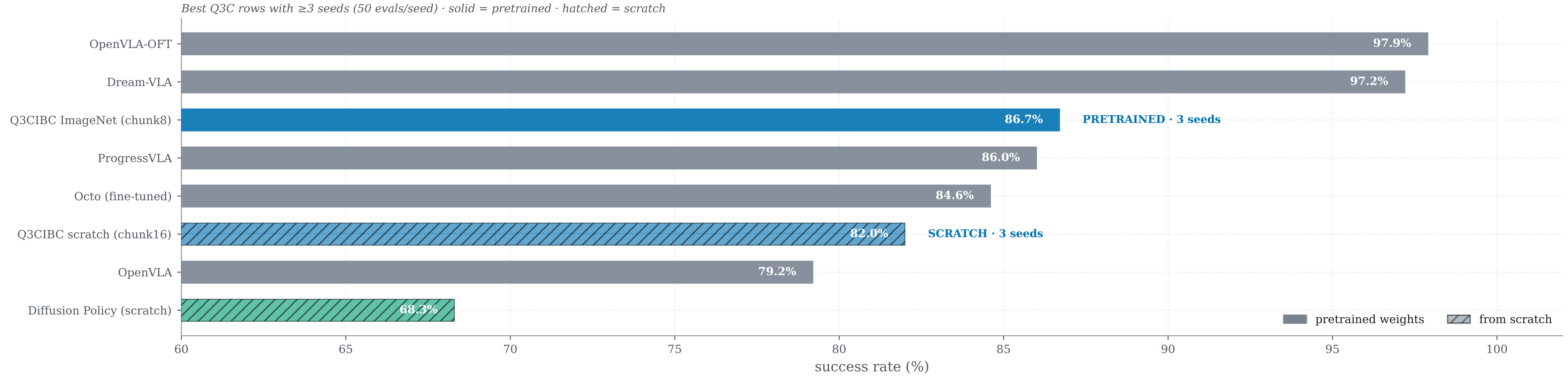
(f) D4RL Pen

Average reward vs. latency; only official IBC quality shown



(g) LIBERO-Goal: standard protocol

Best Q3C rows with ≥3 seeds (50 evals/seed) · solid = pretrained · hatched = scratch



Q3CIBC IBC EBM Explicit BC Diffusion Policy Other

Summary. Q3C traces strong low-latency frontiers; refinement yields its largest gain on Kitchen, while pretrained visual representations improve LIBERO-Goal performance.

Source: final comparison CSVs in results/particle and results/hyperparam_search/combinedv2_cpascounter_training.